

Vehicle Emissions: From (Tail)Pipe Dream to Exhaust Nightmare

In the last chapter, we saw that passenger cars, specifically ICE-powered vehicles, have been the dominant form of personal mobility for the last century. These cars have lifted people out of poverty and provided a level of freedom that is arguably unparalleled by other inventions. As we learned, ICEVs replaced horses to reduce pollution, but we have accidentally replaced one form of pollution with another. Not always, but in general, the emissions associated with building and driving vehicles have a negative impact on our environment and wellbeing. Of course, this is a very general statement, and we'll break it down in more detail in this chapter.

There are two ways we can reduce emissions: (1) use vehicles less and build fewer of them or (2) reduce or eliminate the emissions associated with manufacturing and driving vehicles.

What kinds of emissions are we talking about exactly? We'll take a look at vehicle emissions from first principles in this chapter. Just a warning, parts of this get a bit technical (if you were hoping to never see a chemical equation after your high school chemistry course, you're out of luck), but the technical parts are important for understanding where emissions come from and, thus, how we can reduce them.

Overview of Emissions

Let's split the emissions from internal combustion engines into three categories:

1. Harmless
2. Greenhouse gases
3. Pollutants (those that are harmful to humans in some way if inhaled)

As we've noted, emissions are not unique to ICEVs. Emissions from horses, such as manure, were toxic, produced GHGs, and spread diseases, and any technology that replaces the ICE in the future will almost certainly have emissions associated with it.

Types of Emissions and Where They Come From

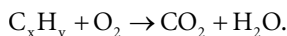
Pollution Is Not a Helpful Word

The word “pollution” is often thrown around when we talk about emissions from vehicles, and we've used it to describe emissions from horses. We all know that pollution is bad, and reducing pollution from cars has long been a priority for the automotive community. But to address our pollution problem, we first must ask, “What exactly *is* pollution?”

To answer, let's consider the journey of your fuel, from when you put it in your vehicle until it leaves the tailpipe. We can start with some simple chemistry and consider our fuel to be a generic hydrocarbon called C_xH_y . A hydrocarbon is an organic compound that includes carbon and hydrogen, and by “ C_xH_y ,” we mean that we have “ x ” atoms of carbon and “ y ” atoms of hydrogen.¹ For example, generic gasoline can be approximated as C_8H_{18} and generic diesel as $C_{12}H_{24}$, although in reality they are complex mixtures containing hundreds of different components.

Ideal Combustion

For ideal combustion, we take our fuel, add exactly the right amount of oxygen to oxidize it, and end up with carbon dioxide and water (well, most likely steam—combustion is hot after all), which looks like this:



Those with a keen eye for chemistry will notice that this equation, and the ones that follow, are not always balanced. The number of variables and complexity makes balancing them rather messy and, for our purposes, unnecessary. Forgive us.

Of the two gaseous emissions produced—carbon dioxide and water—we can put CO_2 comfortably in the GHG category. But how should we categorize water? Water vapor is technically a GHG; it absorbs long-wavelength radiation and radiates it back to the earth's surface, causing warming. In fact, water vapor is the earth's most abundant GHG, making up about 1% of the atmosphere at any given time. However, water vapor does not have a particularly long lifetime, and within days of being emitted into the atmosphere, it typically precipitates out (i.e., it rains). But as the atmosphere warms, its capacity for retaining water vapor increases, and so rising global temperatures may cause a positive feedback loop. That said, transportation and other human sources are not currently adding significant quantities of water vapor to the atmosphere, so its contribution to anthropogenic (human-influenced) climate change is negligible. So, for now, let's put water vapor in the “harmless” category.

¹ Some fuels, including many biofuels and e-fuels (which are often oxygenated), do not contain only carbon and hydrogen atoms, as this example does. The equations for these fuels are a little more complex, but the fundamentals are the same. In addition, hydrogen, which we will discuss in Chapter 8, does not include carbon.

At the end of our ideal combustion, this leaves our emissions count as:

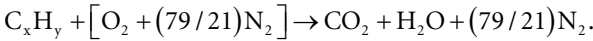
Harmless	Greenhouse gases	Pollutants
Water vapor	Carbon dioxide	

Combustion in Air

Unfortunately, in practice, it is rather difficult to combust fuels in pure oxygen. While not impossible, it would be very inconvenient and expensive to haul around a large oxygen tank on every vehicle. Instead, we go the far more practical route and burn our fuels in air.

Air is a complex mixture of gases. Dry air consists of 78% nitrogen, 21% oxygen, 0.9% argon, 0.04% carbon dioxide (well, 0.04% at the time of writing; when we were in school it was 0.03%, and the fact that this number is rising is a problem), and small amounts of other gases, such as neon, helium, methane, etc. [1]. In addition, there are varying levels of water vapor, commonly accounted for as relative humidity.

As engineers, we approximate all the time, and that's what we're going to do here. For our purposes, we can consider air to be a mixture of only oxygen and nitrogen (for simplicity, we'll include argon and everything else in with nitrogen); so, per mol of oxygen, air can be expressed as: $O_2 + (79/21)N_2$. Now we need to modify our combustion process:

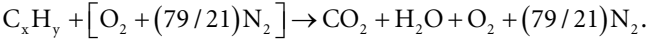


This hasn't changed much compared to the ideal case above—the nitrogen, being largely inert, just sits on both sides of our equation. Nitrogen is pretty uncontroversial, so we can safely add it to our “harmless” list.

At the end of our combustion-in-air process, our emissions count is:

Harmless	Greenhouse gases	Pollutants
Water vapor	Carbon dioxide	
Nitrogen		

So far we've assumed that we have exactly enough air to combust the available fuel. In the jargon, this is called stoichiometric combustion. However, this need not be the case. For example, if we had excess air (lean, in the jargon) we would end up with “spare” oxygen on the right-hand side:



If we had excess fuel (rich, in the jargon), we would end up with “spare” fuel on the right-hand side:



It's possible to chemically balance these equations to work out exactly how much of what is on each side—see Stone's (2012) book for an example [2]. This enables us

to calculate the air:fuel ratio (AFR). Once we have done this, we can define a parameter to express how stoichiometric the combustion is, λ (lambda), or its inverse ϕ (phi),² which is called the equivalence ratio:

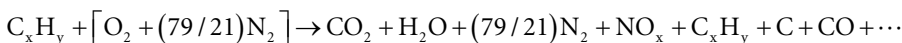
$$\lambda = \frac{\text{AFR}}{\text{Stoichiometric AFR}} = \frac{1}{\phi}.$$

For $\lambda < 1$ ($\phi > 1$), there is excess fuel, and the mixture is rich, and when $\lambda > 1$ ($\phi < 1$) the mixture is lean. We will see conditions when both of these might be advantageous (or indeed detrimental) later on.

Real Combustion Is Hot and Happens Quickly

When you bought this book, you were probably hoping to get some profound and clever insights into the world, so here's one for you: combustion is hot! In internal combustion engines, the combustion temperatures exceed 1500–2500°C. Combustion also needs to happen quickly—very quickly—otherwise your engine won't rotate fast enough to deliver the power required for your vehicle.

These high temperatures (a.k.a. hotness) and fast speeds lead to a number of other emissions from our process: carbon monoxide (CO); oxides of nitrogen (NO_x), mostly NO and NO₂; carbon (C); and other hydrocarbons (C_xH_y), including some unburned fuel. This means we need to modify our combustion process again:



Other emissions can result from this process (noted by the "+..."), and we'll mention a few of these, but NO_x, C_xH_y, C, and CO form the majority of the additional emissions. How do hotness and speed create these emissions?

Hotness

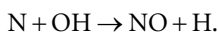
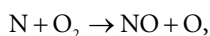
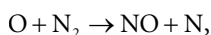
Hotness is the main source of NO_x, which is formed by the reaction of oxygen and nitrogen in the air. In fact, one of the most common natural sources of NO_x formation is a lightning strike. NO_x is actually three main chemical compounds: nitric oxide (NO), nitrogen dioxide (NO₂), and nitrous oxide (N₂O). NO_x can be formed by three different pathways (known as mechanisms), which we have listed in order of importance:

1. Thermal
2. Prompt
3. Nitrous oxide

The *thermal* mechanism is typically governed by a set of chemical kinetic equations called the extended Zeldovich mechanism (named for Russian physicist Yakov Zeldovich). These reactions are mostly dependent on temperature, but also on the

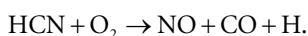
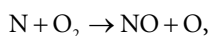
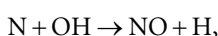
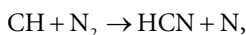
² Yes, it is a bit confusing that both λ (the air-fuel equivalence ratio) and ϕ (the fuel-air equivalence ratio) exist when they are the inverse of each other. It seems that people who work on conducting experiments with engines (Felix) prefer λ , and people who model them (Kelly) prefer ϕ .

local AFR. This mechanism dominates in most ICEs and is comprised of three key chemical balance equations³ [3, 4]:



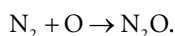
These equations are typically valid above approximately 1500°C.

Prompt NO_x is formed by the following equations [5]:



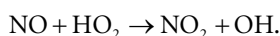
This mechanism is not significantly temperature dependent, which makes it challenging to suppress; thermal NO_x, on the other hand, can be suppressed by reducing combustion temperatures with, for example, exhaust gas recirculation (more on this in Chapter 4). However, because the temperatures in ICEs are so high (combustion is hot, remember?), the thermal mechanism dominates, so the prompt mechanism does not typically contribute much to NO formation in a conventional engine. It can still be significant, though, in some specific circumstances, such as when there is fuel-bound nitrogen or if combustion is occurring at low temperatures.

The *nitrous oxide* mechanism produces N₂O rather than NO and can also be important at lower temperatures and/or when there is significant excess air (lean, λ > 1). But again, this is usually dominated by the thermal mechanism. The nitrous oxide mechanism is [6]:



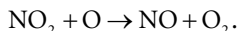
Excess air is important for all three mechanisms. Looking at the equations above, you can see that the presence of oxygen is required for NO_x formation, as you might expect. If an engine is operating stoichiometrically (just enough oxygen to fully combust all of the available fuel), then there is limited oxygen available to form NO_x, because the oxygen will react preferentially with the fuel. Almost all of today's gasoline engines operate stoichiometrically, and almost all diesel engines do not.

Above about 700°C, which internal combustion engine cycles usually exceed, the NO formed by any of the three mechanisms above is quickly converted to NO₂ through the following reaction:



³ Strictly speaking, these equations, as with others later, are balance equations. This means that, depending on the conditions, they could happen either forward or backward. Given that we are discussing NO_x formation, we have put them in a form where NO_x is the result.

Above 1200°C (again, typical for most ICE cycles), this NO₂ is converted back to NO by the following mechanism:



However, this last reaction can be stopped or avoided by extinguishing (or quenching) the flame with a cooler fluid, just like blowing out a candle. The thermal mechanism is the most prominent above 1500°C. If it can be suppressed with a cooler fluid, then there will likely be proportionally more NO₂ given the mechanism above. Neither NO or NO₂ is desirable (and indeed NO transforms into NO₂ over time in the atmosphere), but the difference can be important for exhaust aftertreatment systems, which we'll discuss in the next chapter.

You can find good overviews of NO_x formation in ICEVs in the two engine “bibles”: Stone (2012) [2] and Heywood (1988) [7]. For more recent trends, like the increase in the importance of engine-out NO₂, check out [8–11].

Speed

If NO_x formation is due to high temperatures, then C_xH_y, C, and CO are a result of speed. All of these emissions are formed because of incomplete combustion, which happens for two main reasons:

1. There is not enough time for combustion to fully take place.
2. There is not enough air to combust all of the available fuel.

These two scenarios can be linked. For example, the air and fuel are almost never perfectly mixed in the combustion cylinder when the mixture is burned. It would take more time than is available for them to become fully mixed, so often there are some areas where there is not enough air for the amount of fuel needing to be burned. Other times, there is simply not enough air overall for the amount of fuel needing to be burned.

Combustion happens very quickly in an ICE. It takes less than 1.5 ms in a gasoline (or petrol) vehicle at 6,000 rpm and less than 0.5 ms in a Formula 1 (F1) engine. This short timescale means that it is almost impossible to make sure all of the fuel and oxygen molecules meet, so some of the fuel only burns partially. Similarly, if there is not enough air for the fuel available, not all of the fuel will burn completely. This may sound undesirable, but it can allow for higher engine power, because combustion occurs fastest when there is some excess fuel. In addition, it can help avoid exceeding construction material temperature limits, particularly of the turbocharger, since the fuel has a cooling effect.

Chemically speaking, you can think of C_xH_y, C, and CO as various staging posts along the way from our original fuel (C_xH_y) to our final exhaust product (CO₂). Let's consider each of these emissions in turn.

Unburned Hydrocarbons (C_xH_y). C_xH_y is commonly referred to as unburned hydrocarbons or UHCs. UHCs is a catchall term for the hydrocarbons in the exhaust that have not been fully oxidized into CO or CO₂. Some of the UHCs in the exhaust may be the initial fuel that has not reacted with any oxygen, but the bulk of them will be intermediate combustion species (here “species” refers to molecules and

chemical compounds, not living organisms!). The oxidation of hydrocarbons involves an incredibly complex set of chemical reactions (numbering in the thousands) and tens, if not hundreds or even thousands, of intermediate species. Since combustion happens rapidly, there is not enough time for all of the hydrocarbons to oxidize, and if there is excess fuel, then complete combustion is impossible.

UHCs are frequently split into two categories: methane (CH_4), which is a potent GHG, and non-methane hydrocarbons (NMHC), that is, everything else. The effects of these two groups are different, and they are also regulated differently. Methane is a significant emission from liquefied natural gas (LNG) and compressed natural gas (CNG) vehicles. Two-stroke engines, common to small motorcycles, scooters, mopeds, and garden equipment, can produce very high levels of UHC emissions as well. Gasoline engines, because they mostly operate stoichiometrically and there is never perfect mixing, can also emit notable levels of UHCs. Diesel engines, on the other hand, emit low levels of UHCs from the tailpipe because they typically operate with excess air, so there's plenty to go around to oxidize all of the hydrocarbons (although diesel can emit larger amounts of particulate matter [PM], which is formed from hydrocarbons—see the next section). Again, for those interested in knowing more, Stone (2012) [2] gives a comprehensive overview.

Particularly in warmer climates, UHCs can also be emitted directly from the fuel tank via evaporation. This occurs regardless of whether the vehicle is being driven or not, so regulating the emissions per unit of distance driven wouldn't make much sense. Still, testing does occur for these emissions. As a result, several methods have been developed to prevent them, such as devices that inhibit evaporation by capturing the vapor with activated carbon. Fuel pumps at gas (or petrol) stations also include onboard refueling vapor recovery to prevent UHC vapor from being released as you fill up your car.

Carbon (PM). OK, we're cheating a bit here, but carbon is a simplified term in our chemical equation for the large number of emissions that we clump together as PM, also sometimes called soot. We're also cheating because PM is not only a result of speed, but also requires high temperatures. PM can comprise many different things, and rarely is it a lump of carbon on its own (sorry). Instead, PM is typically carbon particles that have adsorbed hydrocarbons on them (i.e., stuck in a layer around the outside). When emitted from an engine these particles are extremely small, usually less than 200 nm in diameter, and sometimes as small as 5 nm. For comparison, a typical human hair is about 100,000 nm in diameter.

PM is formed on a pathway through a type of UHC called polycyclic aromatic hydrocarbons (PAHs)—without going into too much detail, these are rings of six carbon atoms that are stuck together to make multi-rings, a little bit like the logo of the Olympic Games. These PAHs collide and react to form carbon spherules (as small as 15 nm), which then collide, stick together, and absorb other gas-phase emissions. In addition to these solid particles, a variety of gas-phase species condense and form very small particulates. After these particles have been emitted from the engine, they “age” (by reacting under sunlight) and stick together in the atmosphere quite rapidly.

There are two main metrics used to “count” PM emissions: their number (particle number—PN) and their mass (particle mass—confusingly also PM). A lot of particles are emitted from vehicles, around 10-100 million per cubic centimeter of unfiltered

exhaust. For context, in unpolluted (and un-dusty) air, you might expect to see 10,000 particles per cubic centimeter [12]. Because of the small size of these particles, their mass is tiny. The total number of particles emitted during one mile of driving a modern vehicle weighs only a few milligrams (an average pea weighs 200 mg).

A major challenge when counting particulate emissions is that the particle number is not a conserved quantity. That is, if two particles stick together, they form a single bigger particle. Mass, on the other hand, is conserved. If two particles weighing 1 mg each stick together, you still end up with 2 mg of particles. Given the large numbers of particles in engine exhaust and their propensity to stick together, as well as challenges with vapor phases condensing into liquid particles, measuring particle number is extremely challenging and a great deal of effort has gone into establishing robust, repeatable measurement procedures [13].

Older gasoline engines, such as those fitted with carburetors and port fuel injectors (PFI), emit low levels of PM. Modern gasoline direct injection (GDI) engines are more efficient and thus emit less CO₂, but tend to have poorer air-fuel mixing, which leads to higher PM emissions (particularly particle number emissions). Diesel engines, because of their mixing-controlled combustion, have very high levels of PM emissions, so all modern diesel engines are fitted with filters. Increasingly gasoline engines are also fitted with filters, reducing their PM emissions to even lower levels. Two-stroke engines can have incredibly high levels of PM emissions (you may have seen very smoky exhaust from such engines) not only due to poor mixing, but also because some of the lubricant oil is combusted alongside the fuel. Lubricant oil includes PAHs and so is already a step along the path to PM, leading to high levels of PM from its combustion.

This is a very brief overview of particle emissions, and a lot more can be said about PM formation (this was, in fact, the topic of Felix's PhD thesis). You can find a good summary in Peter Eastwood's book *Particulate Emissions from Vehicles* [14].

Carbon Monoxide (CO). CO is formed when there is not enough oxygen to complete the reaction to CO₂. Its presence in ICE exhaust is dependent on the engine's AFR. If there is excess air, there is almost no CO in the exhaust. On the other hand, if there is "exactly" the right amount of air, which is common in most gasoline engines, then there will be some CO formed due to imperfect mixing. If there is insufficient air, then CO can be formed in large quantities. In two-stroke engines, CO is particularly prevalent because mixing in these engines can be quite poor. Conversely, diesel engines almost always operate with excess air, so they emit low levels of CO.

Other Emissions. So far, we have only discussed the "main" emissions. There are others, of course, but most fall into one of the categories above. One that does not is ammonia (NH₃), which is formed by the reaction of nitrogen (from the air) and hydrogen (from the fuel). Although its presence in automotive exhaust is miniscule [10], it is toxic in large quantities and can form atmospheric particulates given time, so it's worth at least being aware of its existence.

Another emission our analysis hasn't covered is sulfur dioxide (SO₂). Before about the 1970s, SO₂ was a significant emission. Sulfur is a natural component in crude oil, so it was also present in the fuels we put in our vehicles. Since then, however, sulfur has been successfully removed from our fuel, and SO₂ emissions from cars today are extremely low. In the UK, this led to a 95% drop in atmospheric SO₂ from

1970 to 2014. These days, the only major source of transportation SO₂ emissions is from shipping, because the fuels used in that sector still contain high levels of sulfur (although this is starting to be regulated).

The final emission we haven't yet mentioned is lead. Tetra-ethyl lead was a common gasoline additive because it substantially increased efficiency and helped prevent engine knock. Knock happens when the fuel ignites before it is supposed to, leading to a characteristic knocking sound. Knock can be avoided by changing when the spark ignites (making it earlier), but this reduces the engine efficiency. Lead did help mitigate knock problems, but as most of you are probably aware, lead is neurotoxic in accumulated doses, meaning it builds up over time—particularly in the brain—and poisons you. For this reason, lead started to be phased out of gasoline beginning in the 1970s, was mostly gone by 2000, and today is present only in gasoline in Algeria, Iraq, and Yemen [15].

In our analysis, we have only considered pure hydrocarbon fuels—we have not discussed oxygenated fuels, which are common in biofuels and e-fuels. Oxygenated fuels behave similarly to pure hydrocarbon fuels, although notable reductions in PM emissions, and in the case of diesel engines, NO_x emissions, have been reported compared to their pure hydrocarbon counterparts. However, increased levels of ethanol, the most common biofuel substitute for gasoline, lead to higher emission levels of formaldehyde and acetaldehyde, known together as aldehydes (although there are much higher sources of aldehyde emissions today such as tobacco and marijuana smoke, building materials, varnish, and paint). Emission of ethanol, itself a pollutant, will also rise.

The Final Tally

How do we categorize these emissions? Well, most of them are pollutant emissions; however, two of them, methane (CH₄) and nitrous oxide (N₂O), are also potent GHGs, so we will include them in that category. Some of these emissions are not present in significant quantities from today's vehicles, so we will put them in brackets—they are worth bearing in mind, but not critically important at the moment.

So, now that we know that combustion is hot and happens quickly(!), we can make the final update to our emissions count:

Harmless	Greenhouse gases	Pollutants
Water vapor	Carbon dioxide	NO _x (NO + NO ₂)
Nitrogen	{Nitrous oxide (N ₂ O)}	Non-methane hydrocarbons
	{Methane}	Particulate matter*
		Carbon monoxide
		{Ammonia}
		{Sulfur dioxide}
		{Lead}
		{Ethanol}
		{Aldehydes}

* Black carbon, which makes up a significant portion of PM (75–80% for gasoline engines), has high global warming potential, but its contribution is typically minor compared to vehicle GHG emissions, so we will leave PM in the pollutant column.

Non-Exhaust Emissions

All of the emissions we've described so far are exhaust emissions. In other words, they are produced in the engine and come out of your car's tailpipe. But driving your car creates another type of pollution: non-exhaust emissions. If these emissions don't come out of the exhaust, then where do they come from? The answer may surprise you.

Tires

What happens when the rubber meets the road? A very slow disintegration of the tire surface! This is why you need to replace your tires from time to time. Rubber particles of many different sizes are released from the tires, and the small ones can even go airborne. These small particles are classified into two groups based on their size: PM_{10} (particles with a diameter of 10 μm or less) and $PM_{2.5}$ (particles with a diameter of 2.5 μm or less). The number of particles that are released is strongly related to the weight of the vehicle and how it's driven, so heavier cars emit many more particles from their tires (all else being equal). In a discussion with Nick Molden, Founder and CEO of the consultancy Emissions Analytics, we learned that a standard vehicle (with four tires) can emit around 75 mg/km of non-exhaust PM on average under normal driving conditions—about 17 times the European tailpipe limit for the engine! These emissions are a side effect of the friction that allows your tires to grip the road; if your tires didn't emit these particles, your tires couldn't grip. This is why cars with very high grip requirements, such as sports cars or F1 cars, need their tires replaced so frequently. While future rubber compounds may reduce this source of particles, and indeed it is an area of significant research, it's very unlikely that these emissions can be completely eliminated.

Brakes

Your car slows down when pressure is applied to the braking system. This pressure forces the brake pad and disk together. Friction between the two causes erosion of both the pad and the disk, resulting in particles. These particles are typically metallic in composition and they subsequently enter the atmosphere just like other PM. Such particles are around 150 nm in diameter, and hot brakes (heated up from heavy braking, for example) can produce particles as small as 30 nm in diameter. Studies have shown that the harder the braking, the more particles are released (as you might expect), and that there are certain temperature thresholds above which very high levels of particulates are emitted [16].

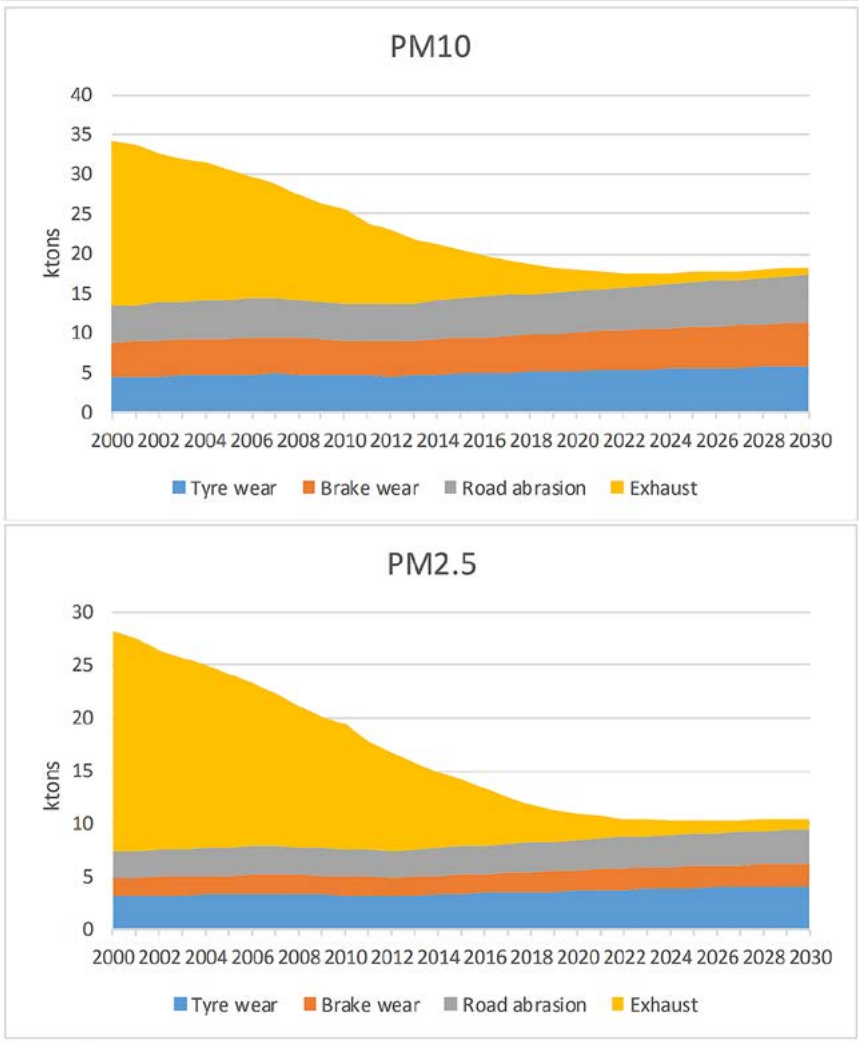
Roads

Just as friction between tires and the road results in the fragmentation of tires, this process also creates particles from the road surface itself (called road abrasion). In addition, dust that gets kicked up by the tires (known as resuspended road dust) adds to the non-exhaust particles.

At this point, you may be thinking, "So what? The real problem is the exhaust!" Well, let's take a look at how PM_{10} and $PM_{2.5}$ from both exhaust and non-exhaust sources have evolved over the years in the UK, as reported by the Air Quality Expert Group [17]. This is shown in [Figure 3.1](#), which includes both data (2000–2016) and

FIGURE 3.1 PM₁₀ emissions (top) and PM_{2.5} emissions (bottom) from road transport in the UK. In these graphs, 2000–2016 is based on actual data, while 2017–2030 is a projection [17].

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projections (2017–2030). While exhaust PM has been significantly reduced since 2000 (due to improved aftertreatment technologies, as will be described later), non-exhaust PM slowly increases every year due to increasing traffic. Today, the PM from tires, brakes, and roads is higher than the PM from the exhaust (see [Figure 3.1](#)). This non-exhaust portion of road transport PM has become the larger problem, and it is independent of what is powering your car. If you didn't know that before reading this book, you're not alone! We will revisit these non-exhaust emissions later.

How Emissions Affect the Environment and Our Health

Now that we know what emissions vehicles produce, let's dive into how the non-harmless emissions individually and collectively impact both the environment and our health.

Greenhouse Gases

Carbon dioxide (CO₂) is a colorless gas and is known to be the primary carbon source for life on earth. CO₂ is also a GHG, and, as such, is partially responsible for maintaining a habitable temperature on earth. Like all good things, though, there can be too much of it.

During the 10,000 years before the industrial revolution, the earth's atmosphere contained between 260 and 280 parts per million per unit volume (ppmv) of CO₂. The highest concentration of CO₂ ever recorded in the earth's atmosphere prior to the industrial revolution was 300 ppmv about 300,000 years ago. As of 2020, the concentration of CO₂ in the earth's atmosphere is around 416 ppmv.

Burning fossil fuels (initially coal, then subsequently oil and gas) has contributed significantly to this rise in atmospheric CO₂ concentration, and the use of gasoline and diesel in ICEs is no exception. While the link between atmospheric CO₂ and global temperatures is complex, the outcome is clear: increased levels of atmospheric CO₂ lead to increased global temperatures. The Intergovernmental Panel on Climate Change has recommended limiting the temperature rise to 1.5°C above pre-industrial levels to avoid significant climate change. It is unclear what the highest acceptable level of atmospheric CO₂ is to keep global temperatures in check (and it likely won't be clear until it is too late), but current estimates set 450 ppmv as the upper limit. What is clear is that we need rapid, urgent action to reduce CO₂ emissions, and this, in large part, is what this book is about: the best path to reach this goal.

Methane (CH₄) is a more potent atmospheric GHG than CO₂, with a 100-year global warming potential (GWP) of 32—in other words, 1 ton of methane in the atmosphere has the same warming impact as 32 tons of CO₂ [18]. However, methane is much shorter-lived in the atmosphere than CO₂ (the 100-year GWP takes this into account). This means that eliminating methane emissions will have a more rapid impact on global temperatures than eliminating CO₂. Even if we suddenly stopped emitting all CO₂, it would still take a long time for the CO₂ already in the atmosphere to dissipate. The methane in the atmosphere, on the other hand, will disperse much quicker and thus have a more rapid impact on the climate.

Methane emissions made up 10% of all GHG emissions in 2018 [19]. Extracting and refining oil and natural gas produces around 20% of all atmospheric methane, but methane emissions from vehicles are not currently a significant source of atmospheric methane [20] (although adding more LNG or CNG vehicles may increase this contribution). Nevertheless, any decrease in methane emissions from vehicles would be beneficial given its high GWP.

Nitrous oxide (N₂O) is an incredibly potent GHG—it has a whopping 100-year GWP of approximately 300 [20]. As noted previously, N₂O is a possible emission from automotive engines, but the levels are currently low, less than 1% of the total NO_x

emissions [21]. Still, given its high GWP, we need to remain vigilant in our efforts to monitor and eliminate it. Nitrous oxide emissions made up 7% of all GHG emissions in 2018 [19]. The automotive sector contributed less than 0.25% of this, and that number is trending downward [19]. Agriculture is by far the largest nitrous oxide emitter globally, producing more than 80% of the total [19].

As we noted above, **black carbon**, a component of PM, also has an exceptionally high 100-year GWP of 430–1,500 (sources vary) [22]. Its overall contribution from modern vehicles is low, particularly as particle filters are fitted to many modern vehicles. However, retrofitting older vehicles that emit high levels of black carbon may reap substantial rewards not only for local air quality but also for GHG reduction.

Overall, it is common to express all of these GHGs in units of g CO₂ equivalent. What this means is the GWP of each of these climate change-causing substances is translated to a common unit based on CO₂—the most common GHG, which made up 81% of all GHG emissions in 2018. Sometimes you will see this expressed as g CO₂e [19].

Before we move on to our next category of emissions, it is worth noting that not all GHGs contribute to the warming of the earth. For example, sulfur dioxide and ammonia have 100-year GWPs of –76 and –15, respectively, which means they act as cooling gases [18]. Unfortunately, they are both harmful to humans. Similarly, NO_x has a 100-year GWP of approximately –150, but, as we will discuss shortly, also has a significant negative effect on human health.

Pollutant Emissions

In 2012, the World Health Organization (WHO) declared diesel exhaust, a major source of PM emissions, to be a carcinogen (specifically causing lung cancer). This puts it in the same category as other carcinogens such as asbestos, smoking tobacco, alcoholic beverages, wood dust, and paint exposure [23]. At the time when the WHO classified diesel exhaust, there was insufficient evidence to comment on gasoline exhaust. It is worth noting that the diesel exhaust of 2012 is not the same as diesel exhaust today—the widespread introduction of diesel particulate filters and selective catalytic reduction (discussed in detail in the next chapter) have made a substantial difference. With that broad context from the WHO, let's dig into the details.

Pollutant emissions contribute to poor air quality globally. When we say “poor air quality,” we mean air that contains compounds that can be harmful to human health, depending on the quantity. Exposure to small amounts of any pollutant is unlikely to do us significant harm, but exposure to large quantities of a pollutant can have short- or long-term negative consequences. We also need to be mindful of cumulative effects—if you're exposed to low levels of certain pollutants for a long period of time, they can build up and cause considerable problems. One of the most harmful forms of air pollution is smog.

There are two types of smog: photochemical and smoke-derived “classic” smog. Classic smog episodes were first widely observed in London beginning in the 19th century and continued through the middle of the 20th century. These smog episodes were primarily associated with emissions from industry and domestic coal burning, as classic smog is caused mainly by the smoke and PM from burning coal, which has a high sulfur content. Today, smog can also be caused by agricultural burning, for example, in parts of India.

First observed in Los Angeles in the 1950s, photochemical smog is a severe, visible air pollution episode that is commonly associated with automotive emissions. Rather than being a direct consequence of emissions, photochemical smog is caused by chemical reactions of emissions in the atmosphere triggered by sunlight. Photochemical smog does not describe a specific mixture of chemical components, but rather a variety of mixtures depending on the composition of emissions and the atmospheric conditions or weather. Generally speaking, smog is toxic to humans and can cause severe sickness and premature death.

Ozone (O_3) is a mixed bag for humans. High up in the atmosphere (specifically the stratosphere), it protects us from the harmful ultraviolet radiation from the sun, which is why chlorofluorocarbons (CFCs) destroying the ozone layer was so concerning. Ground-level ozone (in the troposphere) is a different story. Ozone is the main component of photochemical smog. It is extremely reactive and causes lung irritation, chest pain, and coughing. Ozone can affect the respiratory system of otherwise healthy people and more severely affect those with existing conditions, such as asthma or chronic obstructive pulmonary disease (COPD), as well as children and the elderly [24].

There is considerable uncertainty both about how damaging air pollution is to human health and what the main causes are. However, we can still draw some useful conclusions. For example, there remains some uncertainty about the exact mechanisms by which tobacco smoke leads to lung cancer, but the vast majority of experts still agree that smoking causes lung cancer. In general, exposure to PM is the most harmful, followed by ozone and NO_2 . There is an excellent overview from the Winton Centre at the University of Cambridge of the science of air pollution's effect on humans that also grapples with the uncertainty [25].

One way to think about it is that air pollution effectively accelerates aging. We are all going to die; it is merely a question of when. There is a growing, although not yet comprehensive, body of literature suggesting that exposure to air pollution brings our deaths forward [26, 27]. If we have a medical condition already or are in another high-risk category, then every unit of pollution we are exposed to effectively shortens our lives. We find this to be a more helpful way of thinking about it, rather than quantifying the number of people killed every year by pollution, because air pollution rarely appears as the direct cause of death.

So what does this all mean? Obviously, air pollution is bad. It does not affect all of us equally, but it is in everyone's best interest to reduce it as much as possible. Let's now take the list of pollutants from our combustion equations and see how each of them contributes to air pollution.

NO_2 is one of the two main contributors to the formation of smog and ground-level ozone, but that's not all—it also leads to acid rain, causes asthma, directly worsens respiratory diseases (e.g., bronchitis and emphysema), and aggravates existing heart conditions. As we discussed earlier in this chapter, the majority of automotive NO_x is NO, but subsequent oxidation leads to NO_2 in the atmosphere. It is this NO_2 that reacts with non-methane hydrocarbons in the presence of UV light (from the sun) to form photochemical smog.

Unburned hydrocarbons are the other main contributor to the formation of photochemical smog and ozone. There is an extraordinarily wide range of UHCs

possible, and in air quality circles, these are often termed volatile organic compounds (VOCs). VOCs themselves can be very toxic (as an extreme example, we wouldn't recommend drinking gasoline or diesel) and inhalation in sufficient quantities can lead to respiratory, allergic, and immunological effects. In addition, some VOCs are carcinogenic—benzene (C_6H_6) is the most obvious example, and its emission from vehicles is closely monitored and controlled.

Particulate matter is the most harmful of all the direct pollution emitted from a vehicle. We have long understood there is a link between PM and lung cancer, mainly as a result of research into smoking cigarettes. PAHs adsorbed onto the particles can bond covalently with DNA, RNA, and proteins, and when the resultant products are processed by one of the body's organs, this can promote cancerous growth in that organ.

Most of the interest in and regulation of PM emissions to date has been focused on the mass of particles emitted. Similarly, monitoring and legislation of atmospheric air quality have focused on the mass of particles of a specific size. Particles with a diameter of 10 μm or less (PM_{10}) can be inhaled into the lungs; still finer particles, with a diameter of 2.5 μm or less ($PM_{2.5}$), can penetrate even farther. Both of these are regulated by particle mass. However, recent studies have shown that the health effects of particles seem to be dependent on the number of particles and their size rather than the mass. The smaller the particle, the more dangerous—it has been shown that ultrafine particles (defined as smaller than 100 nm) can be passed into the human bloodstream through the alveoli in the lungs within a minute of inhalation [28].

The harmful effects of particulates are thought to be caused not only by the particulates themselves, but also the components that are adsorbed onto them. Both increase the levels of free radicals and reactive oxidation species, which leads to an increase in oxidant species relative to antioxidants. This is called oxidative stress and is thought to be the pathology of many diseases [14].

It is clear that PM emissions are harmful to human health. There is strong evidence suggesting that PM emissions increase the risk of heart attack, stroke, lung and other types of cancer, asthma, COPD, bronchitis, dementia, and Alzheimer's disease. While the precise effects and mechanisms are not yet clear, efforts to reduce particulate emissions will reap significant health rewards for the population.

Carbon monoxide (CO) is a toxic gas. The hemoglobin in our blood preferentially bonds with carbon monoxide over oxygen, and so, in high doses, carbon monoxide can lead to death. You might have a CO detector in your house for this very reason, because a major source of high CO levels is faulty domestic boilers. However, the amount of carbon monoxide in the atmosphere is almost never anywhere near this level. Carbon monoxide has another effect, though: it plays a major role in the formation of ground-level ozone and photochemical smog [29]. This is the main reason for our concern about its presence in automotive exhaust.

Ammonia (NH_3) is toxic in high concentrations, but its main role in atmospheric pollution is the formation of secondary particulates (reacting particularly with NO_2 and SO_2). Ammonia emissions, which today are mostly from farming, constitute a substantial fraction of $PM_{2.5}$ aerosols and thus form a component of smog.

Sulfur dioxide (SO₂) can cause irritation to the nose, eyes, throat, and lungs. It is also a major component of acid rain and reacts with ammonia, and other substances, to form smog.

Lead is happily now just a footnote in most of the world, but 50 years ago was probably the biggest health concern in terms of vehicle pollution. Lead is particularly dangerous because it builds up in the body, and exposure to lead can cause anemia, kidney damage, and brain damage. Very high lead exposure can even cause death.

Ethanol emissions today come mainly from manufacturing spirits (whisky, vodka, gin; pick your favorite!), but in a future with widespread biofuel use, there may be significant ethanol emissions from vehicles. Ethanol emissions in the atmosphere promote peroxyacetyl nitrate formation, a component of photochemical smog [30]. In addition, ethanol emissions decay into aldehydes in the atmosphere.

Aldehydes can cause irritation to the eyes, nose, and throat, notably resulting in asthma-like symptoms. Both formaldehyde and acetaldehyde (the two most common aldehydes emitted) are considered carcinogenic [31, 32], and to really bring it home, acetaldehyde is a contributing cause of hangovers after alcohol consumption!

The Last Mile

So, to summarize this chapter, what do we mean by pollution? Carbon dioxide? Carbon monoxide? Methane? Particulate matter? Nitrous oxide? NO_x? Non-methane hydrocarbons? Ethanol? Ammonia? Aldehydes? Sulfur dioxide? Lead? Something else...? All of these are pollution in their own way. Different vehicle technologies have advantages and disadvantages with regard to each type of pollution and different approaches to reducing and eliminating them.

This blanket term inevitably masks important differences, so when we say we want to reduce pollution, we must specify what. All vehicles, whether gasoline, diesel, or electric, emit pollution—some at the source, others elsewhere. A simple “reduce pollution” is insufficient; we must be specific.

Emissions from vehicles are a big problem when left unchecked. We learned what emissions there are and how they are formed, and we unpacked their impact on our health and our planet’s health. But—before you feel too guilty about driving your car, you should know that we have made great progress in reducing the harmful emissions that come out of its tailpipe (the exhaust contributions to PM shown in [Figure 3.1](#) highlight this well). This has happened, in large part, through government regulations, which are the focus of the following chapter.